TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	78.7 ± 2.79
Correlation Stability	86.2 ± 3.28
Distribution Consistency	85.9 ± 2.43
Entropy Preservation	88.9 ± 1.70
Mean Z Score Drift	0.2 ± 0.04
Categorical Fidelity	83.1 ± 0.54
Privacy Risk	49.6 ± 0.80
Data Density	99.9 ± 0.14
Coverage Score	20.2 ± 5.91
Feature Accuracy	79.0 ± 3.87
Novelty Score	0.5 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	42.865	-1.881	4.2%	OK
TB	3.299	2.459	-0.840	25.5%	Drift
DB	1.486	1.054	-0.432	29.1%	Drift
Alkphos	290.576	263.712	-26.864	9.2%	OK
Sgpt	80.714	50.200	-30.514	37.8%	Drift
Sgot	109.911	53.858	-56.052	51.0%	Drift
TP	6.483	6.781	+0.298	4.6%	OK
ALB	3.142	3.400	+0.258	8.2%	OK
A/G Ratio	0.947	0.949	+0.002	0.2%	OK
Selector	1.286	1.298	+0.012	0.9%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	78.7 ± 2.79
Correlation Stability (*2)	86.2 ± 3.28
Distribution Consistency (*3)	85.9 ± 2.43
Entropy Preservation (*4)	88.9 ± 1.70
Mean Z Score Drift (*5)	0.2 ± 0.04
Categorical Fidelity (*6)	83.1 ± 0.54
Privacy Risk (*7)	49.6 ± 0.80
Data Density (*8)	99.9 ± 0.14
Coverage Score (*9)	20.2 ± 5.91

Feature Accuracy (*10)	79.0 ± 3.87
Novelty Score (*11)	0.5 ± 0.01

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.6094017094017095	0.6965811965811965	-0.00940170940170939	
F1	0.8102957169257147	0.7021304195271175	0.8042624626888744	-0.006033254236840368 5	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.5632478632478632	0.6905982905982906	-0.01538461538461538 5	
F1	0.8128002697439015	0.653371247418358	0.7918488207225811	-0.020951449021320345	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6128205128205129	0.7	-0.00256410256410257	
F1	0.7997903571549948	0.7105319508589819	0.7963595180246938	-0.003430839130300844 5	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6871794871794872	0.5726495726495726	0.6871794871794872	-3.3306690738754695e -17	
F1	0.7993932122395182	0.6651407285524861	0.7951325047803339	-0.004260707459184399	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.6743589743589744	0.7068376068376068	0	
F1	0.8279091624529537	0.7975181397063789	0.8279091624529537	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.5905982905982906	0.6888888888888889	0.014529914529914523	
F1	0.7821343746683586	0.6951445469676725	0.7935760454701357	0.011441670801777038	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6606837606837608	0.5452991452991454	0.647008547008547	-0.01367521367521366	
F1	0.7643507765597449	0.6367996658514299	0.7532557776434172	-0.011094998916327658	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6162393162393163	0.6700854700854701	-0.016239316239316248	
F1	0.7971299319768493	0.7379269166166071	0.7894072199778259	-0.007722711999023446	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6829059829059829	0.576068376068376	0.6974358974358974	0.014529914529914544	
F1	0.78957358710707	0.6743389860717028	0.7979384207034028	0.008364833596332766	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5589743589743594	0.5683760683760684	0.5675213675213675	0.008547008547008555	
F1	0.5587432119380917	0.6381657925675406	0.5746183196417283	0.01587510770363653	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Male	3.7	2.2	216	179	232	7.8	4.5	1.3
2	48	Male	0.7	0.1	1630	74	149	5.3	2.0	0.6
3	33	Male	2.0	1.0	258	194	152	5.4	3.0	1.25
4	45	Female	1.0	0.3	250	48	44	8.6	4.3	1.0
5	54	Female	1.4	0.7	195	36	16	7.9	3.7	0.9
6	61	Male	0.8	0.2	163	18	19	6.3	2.8	0.8
7	29	Female	1.2	0.5	210	45	50	6.5	3.2	0.8
8	37	Male	0.9	0.3	180	30	40	7.0	3.5	1.0
9	52	Female	2.5	1.0	300	60	70	8.0	4.0	0.9
10	44	Male	3.1	1.5	250	55	65	7.5	3.8	1.1
11	60	Female	1.8	0.7	220	50	60	6.8	3.4	0.85
12	34	Male	0.6	0.2	190	40	45	6.2	3.0	1.2
13	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
14	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
15	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
16	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	49	Male	3.9	2.1	189	65	181	6.9	3.0	0.7
2	42	Male	6.8	3.2	630	25	47	6.1	2.3	0.6
3	56	Male	1.0	0.3	195	22	28	5.8	2.6	0.8
4	34	Male	8.7	4.0	298	58	138	5.8	2.4	0.7
5	45	Male	0.6	0.1	196	29	30	5.8	2.9	1.0
6	75	Male	0.9	0.2	206	44	33	6.2	2.9	0.8
7	29	Female	1.2	0.5	210	35	45	6.0	3.2	0.8
8	54	Male	2.5	1.0	300	50	60	7.0	3.5	0.9
9	38	Female	0.8	0.3	180	25	30	5.5	2.8	0.7
10	45	Male	3.1	1.5	250	40	55	6.8	3.0	0.9
11	62	Female	1.5	0.7	220	30	35	6.2	3.1	0.8
12	50	Male	4.0	2.0	400	60	70	7.5	3.8	1.0
13	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
14	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
15	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
16	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	74	Male	0.6	0.1	272	24	98	5.0	2.0	0.6
2	32	Male	15.0	8.2	289	58	80	5.3	2.2	0.7
3	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
4	39	Male	1.9	0.9	180	42	62	7.4	4.3	1.38
5	64	Male	1.1	0.5	145	20	24	5.5	3.2	1.39
6	48	Male	2.4	1.1	554	141	73	7.5	3.6	0.9
7	29	Female	0.5	0.1	150	22	30	6.0	3.5	0.58
8	45	Male	2.0	0.8	200	40	50	6.5	3.0	0.75
9	38	Female	1.2	0.4	180	35	45	7.0	3.8	0.79
10	52	Male	3.5	1.5	250	60	70	7.2	4.0	0.95
11	60	Female	0.9	0.3	220	30	40	6.8	3.6	0.83
12	34	Male	1.5	0.7	190	45	55	6.7	3.4	0.78
13	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
14	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
15	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
16	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	50	Male	0.9	0.2	202	20	26	7.2	4.5	1.66
2	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62
3	60	Female	1.4	0.7	159	10	12	4.9	2.5	1.0
4	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
5	32	Male	0.9	0.3	462	70	82	6.2	3.1	1.0
6	35	Male	0.7	0.2	198	42	30	6.8	3.4	1.0
7	29	Female	1.2	0.5	210	30	40	6.5	3.2	1.02
8	54	Male	2.5	1.1	300	45	60	7.0	3.8	1.05
9	37	Female	0.9	0.3	180	25	30	5.8	2.9	1.07
10	45	Male	3.1	1.5	250	50	70	6.2	3.5	0.93
11	62	Female	1.8	0.9	220	35	45	6.0	3.0	1.0
12	34	Male	0.7	0.2	190	40	50	7.5	4.0	1.25
13	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
14	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
15	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
16	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	42	Male	16.4	8.9	245	56	87	5.4	2.0	0.5
2	62	Male	1.2	0.4	195	38	54	6.3	3.8	1.5
3	46	Female	14.2	7.8	374	38	77	4.3	2.0	0.8
4	65	Male	0.8	0.2	201	18	22	5.4	2.9	1.1
5	45	Female	0.7	0.2	153	41	42	4.5	2.2	0.9
6	72	Male	0.8	0.2	148	23	35	6.0	3.0	1.0
7	29	Female	0.9	0.3	210	22	30	6.5	3.1	0.8
8	34	Male	1.2	0.4	180	35	40	6.0	2.8	0.7
9	50	Female	2.5	1.0	250	45	55	7.2	3.5	0.9
10	41	Male	0.8	0.2	190	30	35	6.8	3.0	1.0
11	55	Female	1.5	0.5	220	40	50	6.7	3.2	0.95
12	38	Male	0.7	0.2	175	25	28	5.9	2.7	1.1
13	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
14	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
15	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
16	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
2	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
3	26	Female	0.6	0.2	142	12	32	5.7	2.4	0.75
4	38	Male	3.1	1.6	253	80	406	6.8	3.9	1.3
5	49	Male	3.9	2.1	189	65	181	6.9	3.0	0.7
6	31	Male	0.9	0.2	518	189	17	5.3	2.3	0.7
7	34	Female	1.0	0.3	210	45	30	6.5	3.2	0.8
8	29	Male	0.9	0.2	175	40	25	7.0	3.5	0.9
9	52	Female	2.5	1.0	300	50	60	7.2	4.0	1.0
10	41	Male	1.2	0.5	220	55	35	6.8	3.6	0.9
11	37	Female	0.8	0.2	190	30	20	6.0	2.8	0.7
12	45	Male	3.0	1.5	250	70	80	7.5	3.9	1.1
13	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
14	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
15	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
16	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	58	Male	1.7	0.8	188	60	84	5.9	3.5	1.4
2	60	Male	2.3	0.6	272	79	51	6.6	3.5	1.1
3	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
4	24	Female	0.7	0.2	188	11	10	5.5	2.3	0.71
5	58	Female	2.4	1.1	915	60	142	4.7	1.8	0.6
6	36	Male	0.8	0.2	158	29	39	6.0	2.2	0.5
7	34	Female	1.2	0.5	210	45	60	6.5	3.2	0.8
8	29	Male	0.9	0.3	180	25	30	5.8	2.9	0.7
9	52	Female	2.5	1.0	300	70	80	7.0	3.8	1.1
10	41	Male	0.6	0.2	150	35	40	6.2	3.0	0.9
11	38	Female	1.8	0.9	220	50	55	6.8	3.4	1.0
12	45	Male	3.0	1.5	400	90	100	7.5	4.0	1.2
13	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
14	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
15	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
16	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	66	Male	0.7	0.2	239	27	26	6.3	3.7	1.4
2	75	Male	0.9	0.2	162	25	20	6.9	3.7	1.1
3	42	Female	0.9	0.2	165	26	29	8.5	4.4	1.0
4	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
5	19	Male	1.4	0.8	178	13	26	8.0	4.6	1.3
6	66	Male	17.3	8.5	388	173	367	7.8	2.6	0.5
7	29	Female	1.2	0.4	210	22	25	6.5	3.2	1.0
8	37	Male	0.8	0.3	180	30	35	7.0	3.5	1.1
9	54	Female	2.5	1.0	300	40	45	6.8	3.9	1.2
10	45	Male	1.0	0.5	220	25	30	7.2	3.6	1.1
11	62	Female	0.9	0.2	250	20	28	6.0	3.0	1.0
12	39	Male	1.5	0.7	190	35	40	7.5	4.0	1.3
13	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
14	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
15	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
16	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	17	Male	0.9	0.2	279	40	46	7.3	4.0	1.2
2	51	Male	4.0	2.5	275	382	330	7.5	4.0	1.1
3	51	Male	0.8	0.2	175	48	22	8.1	4.6	1.3
4	75	Male	8.0	4.6	386	30	25	5.5	1.8	0.48
5	57	Male	1.0	0.3	187	19	23	5.2	2.9	1.2
6	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
7	22	Female	0.5	0.1	150	25	30	6.5	3.2	1.0
8	34	Male	1.2	0.4	200	40	50	7.0	4.0	1.1
9	29	Female	0.8	0.2	180	30	35	6.8	3.5	1.2
10	45	Male	2.5	1.0	250	60	70	7.5	4.5	1.3
11	39	Female	1.0	0.3	160	28	32	6.7	3.8	1.1
12	55	Male	3.0	1.5	300	80	90	8.0	4.8	1.4
13	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
14	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
15	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
16	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1
2	22	Female	2.2	1.0	215	159	51	5.5	2.5	0.8
3	45	Male	2.3	1.3	282	132	368	7.3	4.0	1.2
4	46	Male	0.6	0.2	115	14	11	6.9	3.4	0.9
5	56	Female	0.7	0.1	145	26	23	7.0	4.0	1.3
6	60	Male	2.0	1.1	664	52	104	6.0	2.1	0.53
7	29	Female	1.5	0.6	210	45	30	6.8	3.5	1.0
8	38	Male	0.9	0.3	150	20	25	7.2	3.8	0.9
9	54	Female	2.1	1.0	180	50	40	6.5	3.0	1.2
10	41	Male	1.8	0.7	220	60	35	7.0	3.4	1.1
11	27	Female	0.5	0.1	160	30	20	6.2	3.1	1.0
12	34	Male	3.0	1.5	250	70	50	7.5	4.0	0.8
13	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
14	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
15	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
16	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3